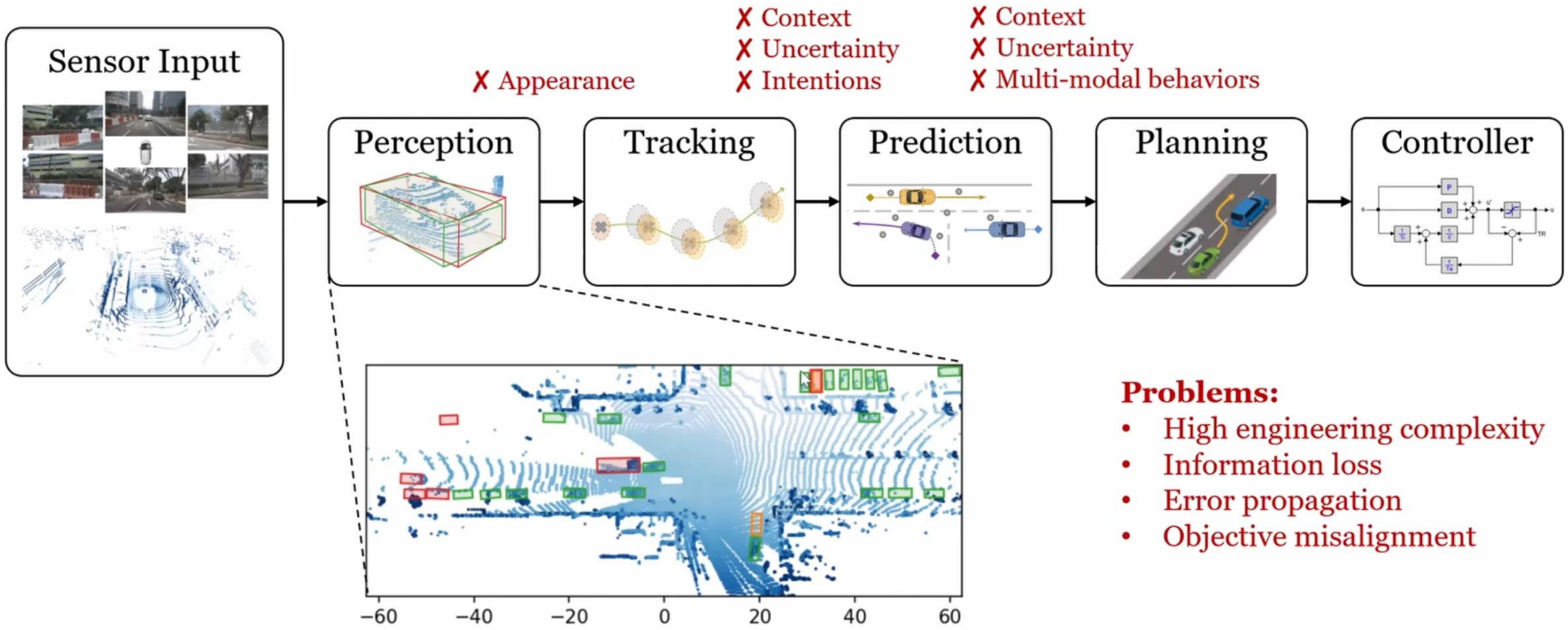
香港城市大學
City University of Hong Kong
Innovating into the Future

Chapter 6: E2E ADS

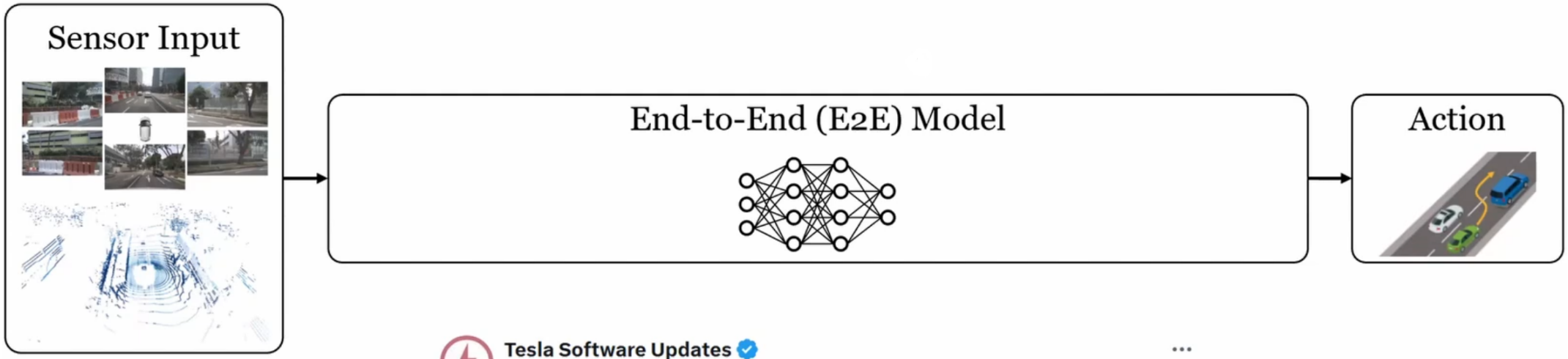
Traditional Autonomous Driving



Problems:

- High engineering complexity
- Information loss
- Error propagation
- Objective misalignment

End-to-end Autonomous Driving

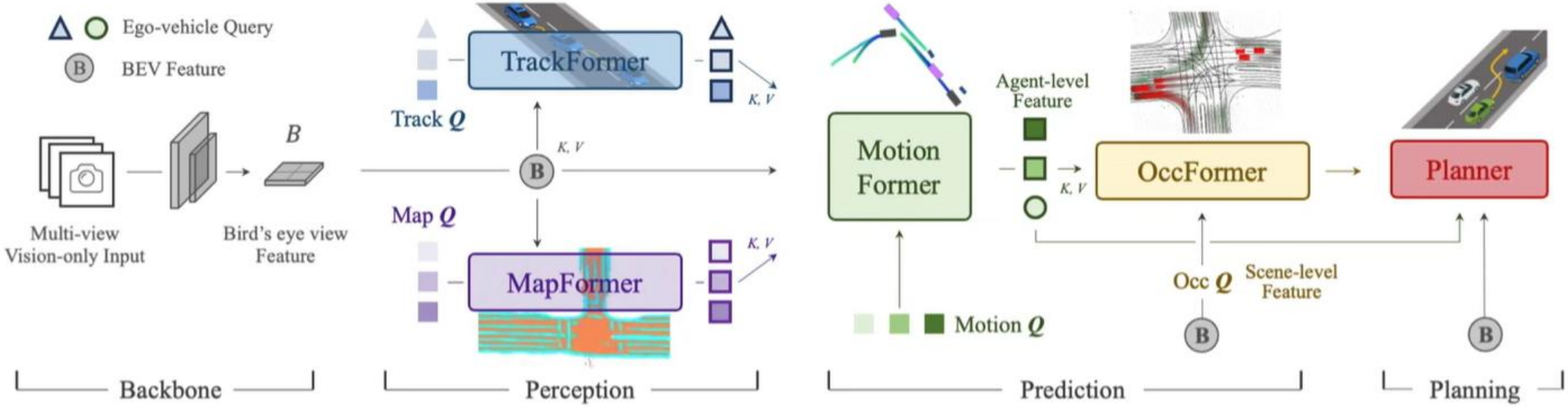


Tesla Software Updates
@TeslaSoftUpdate

🎄Tesla started a new software update 2023.44.30.10 (FSD Beta v12.1)

FSD Beta v12 upgrades the city-streets driving stack to a single end-to-end neural network trained on millions of video clips, replacing over 300k lines of explicit C++ code.

UniAD



Yihan Hu etc. Planning-oriented Autonomous Driving, CVPR 2023 Best Paper Award

UniAD

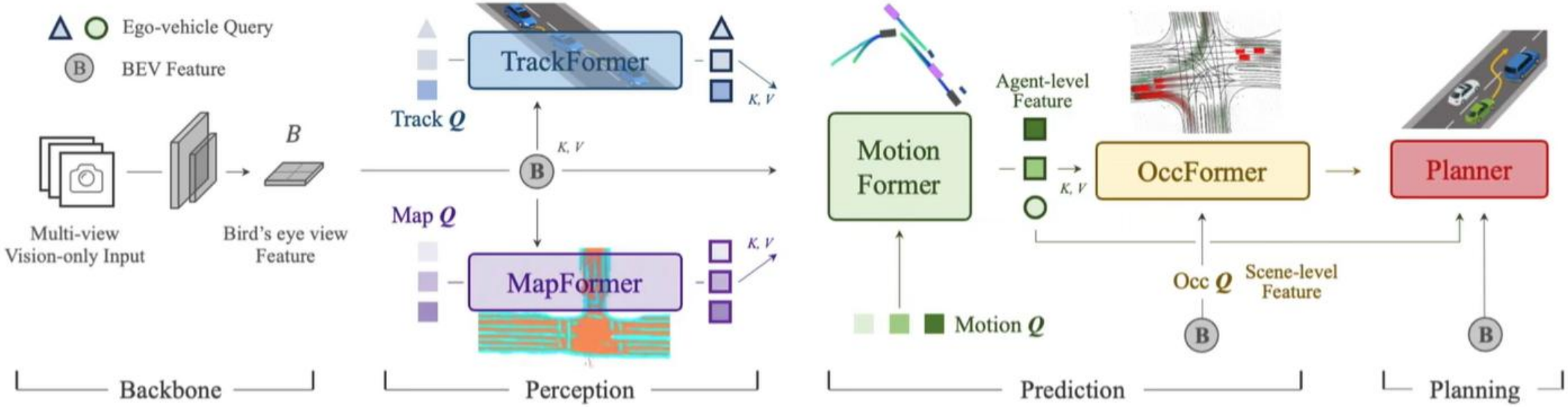
- Track Q : one query for **one agent**
- Map Q : one query for **one map element**

Unified Query

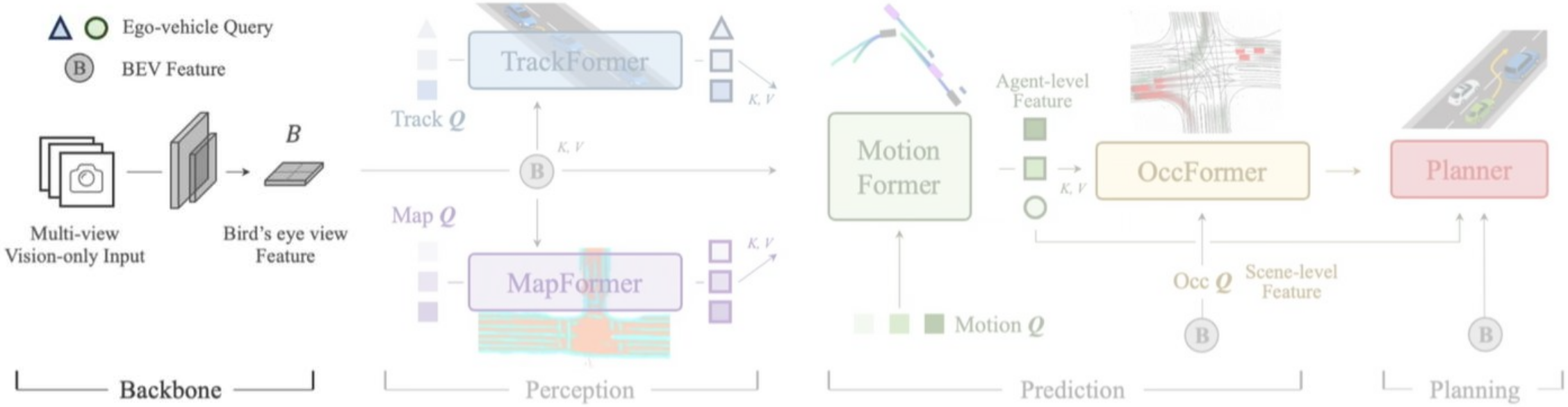
- Motion Q : one query for **one trajectory**
- Occ Q : one query for **one BEV grid**

Each tasks module is a **transformer-based** structure, with:

- ***Attention*** mechanisms model complex relations in the scene
- ***Queries*** interact between tasks modules and transmit upstream knowledge to final planner.

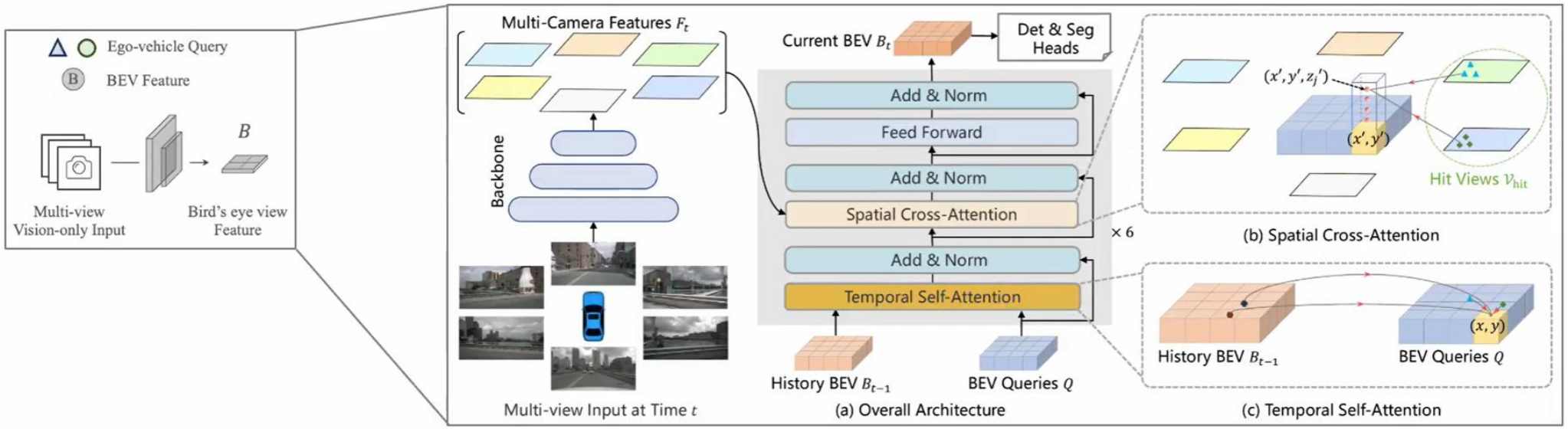


UniAD

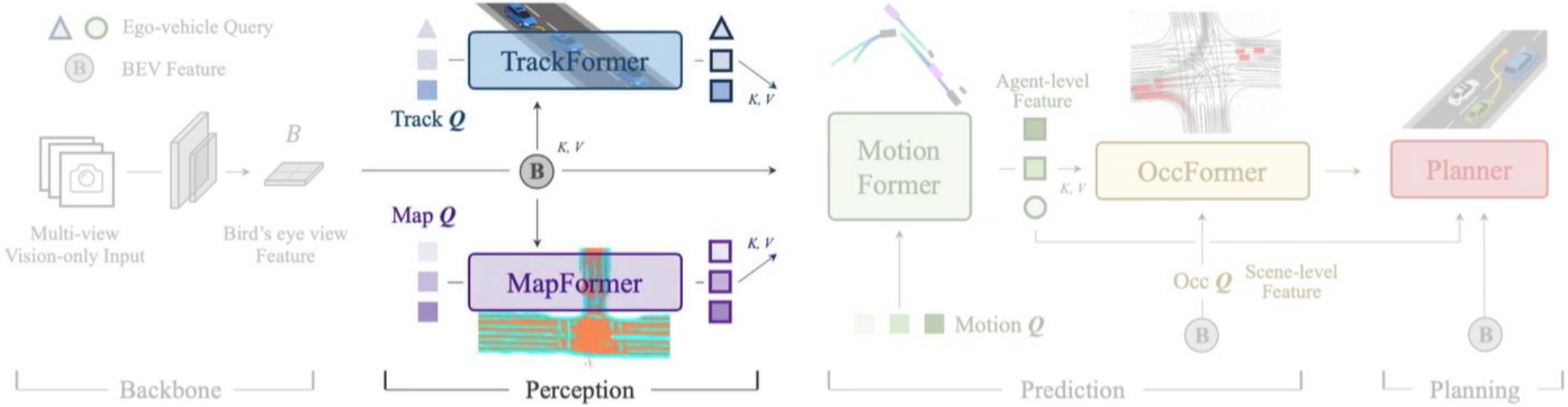

Ego-vehicle Query
BEV Feature
B
Multi-view
Vision-only Input
Bird's eye view
Feature
Backbone
TrackFormer
Track Q
K, V
B
Map Q
MapFormer
Perception
Motion
Former
Motion Q
Agent-level
Feature
OccFormer
Occ Q
Scene-level
Feature
Prediction
Planner
Planning

UniAD

- BEVFormer: a strong BEV encoder for spatial and temporal feature extraction
- You can easily replace it with other advanced BEV encoders



UniAD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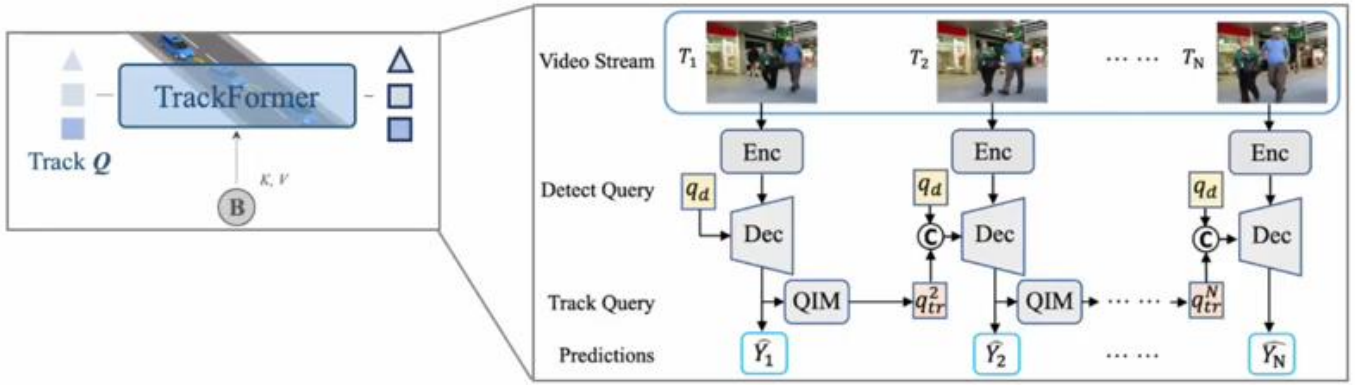
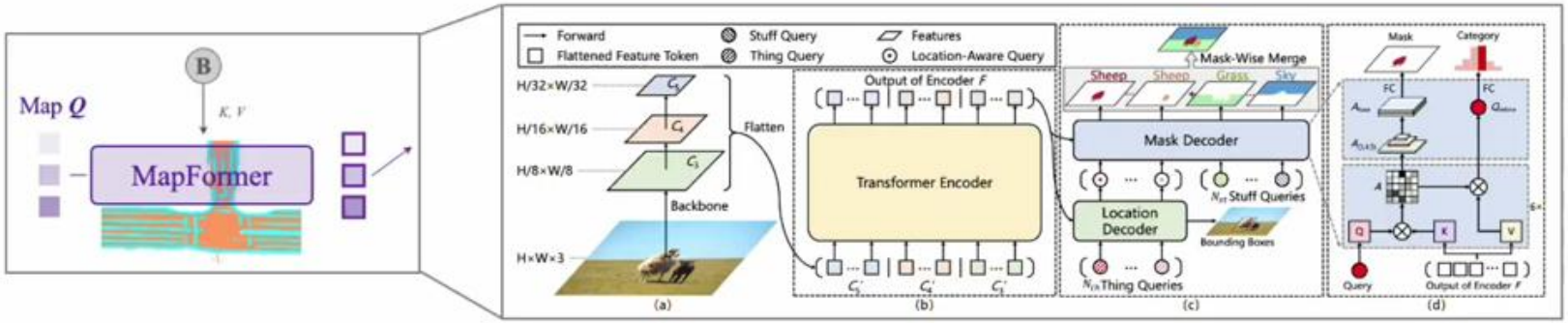
UniAD

TrackFormer



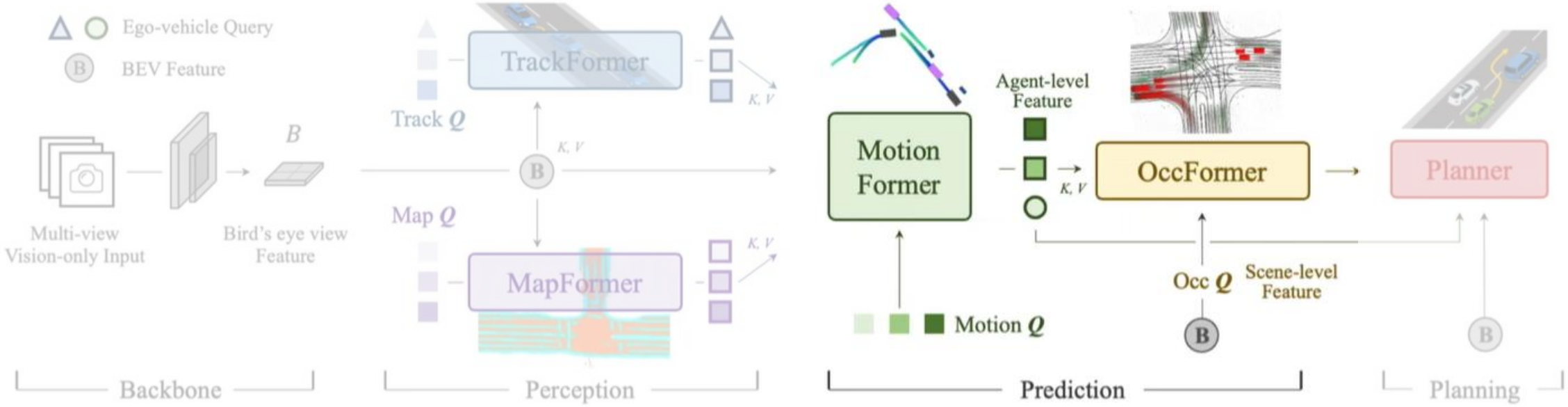
- **End-to-end trainable** tracking without post-association

MapFormer



- Each query represents a map element

UniAD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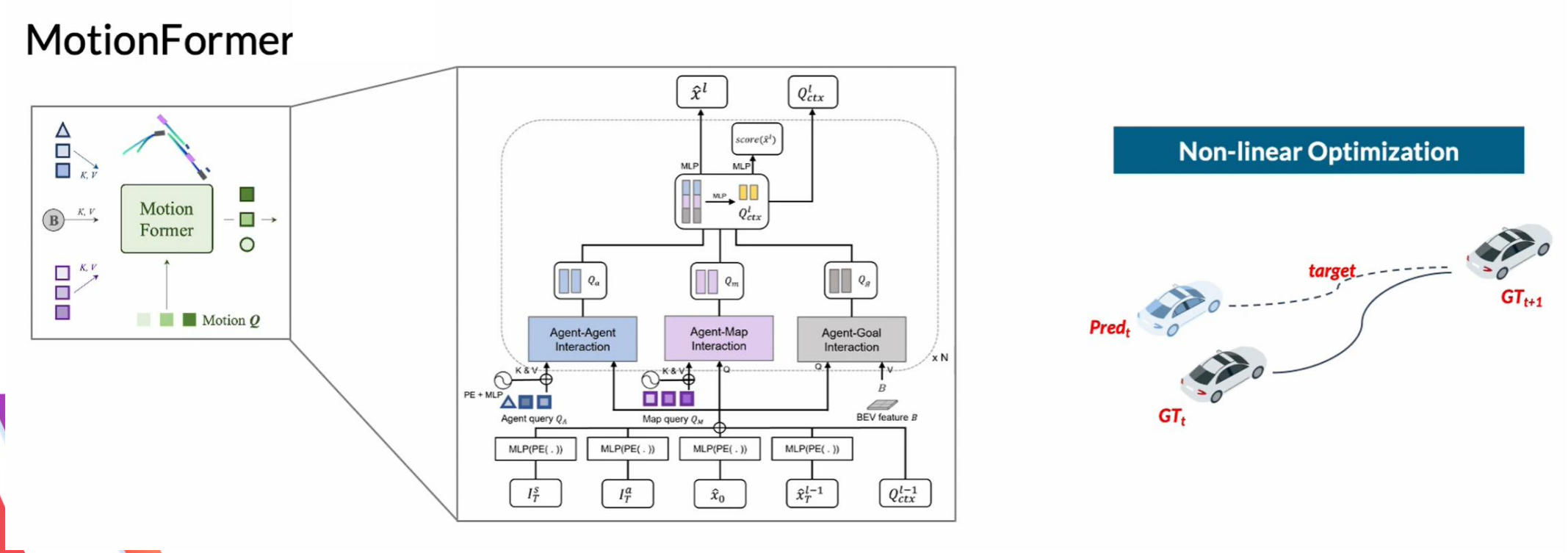
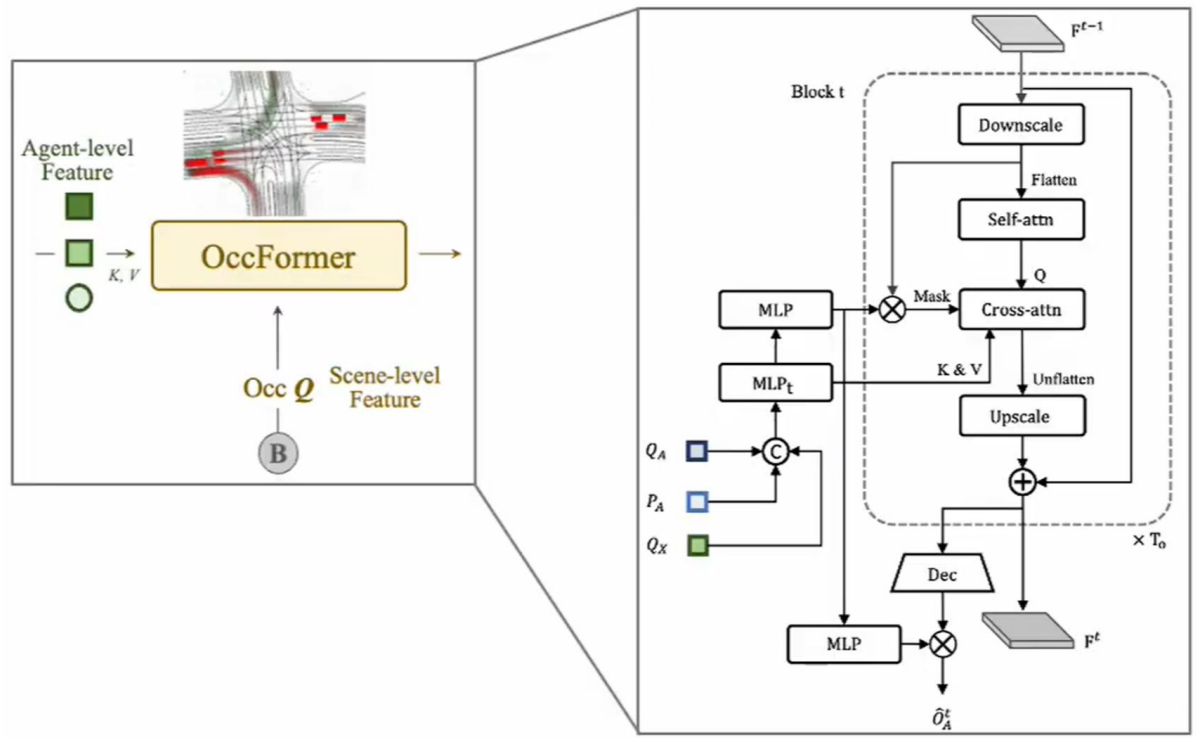
UniAD

- Various relation modeling via attentions:
  - Agent-agent, agent-map, agent-goal point
- Non-linear optimization:
  - Adjust the ground-truth trajectory according to upstream predictions

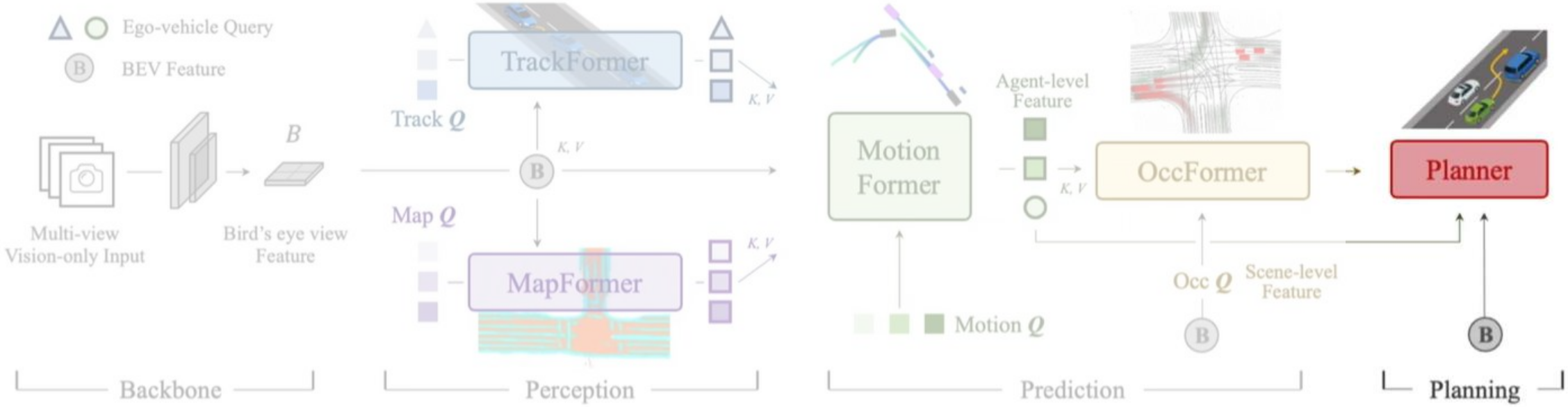


UniAD

- Encode agent-wise knowledge into the scene representation
- Predicted occupancy as attention mask for concentrating the interactions between the agents and their corresponding BEV features



UniAD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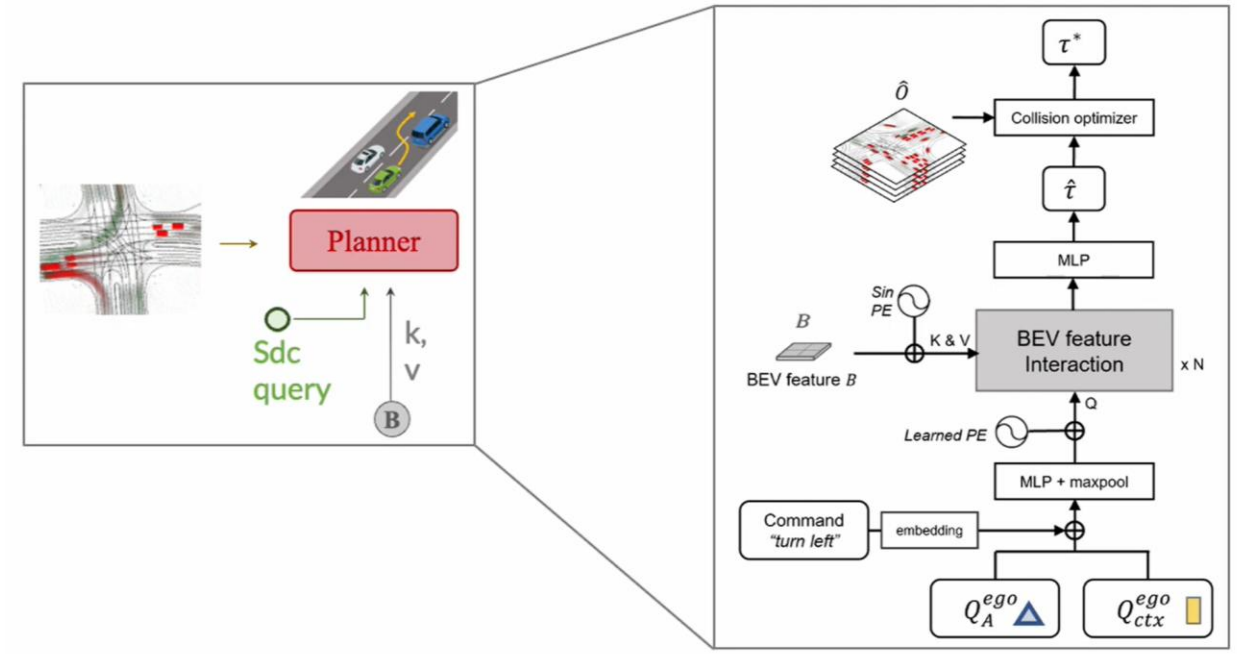
UniAD

- Sdc query: consistently modeling self-car in TrackFormer and Motion
- Formerand is passed to PlannerCollision optimization: Steer the predicted trajectories clear of occupiedareas to avoid potential collisions



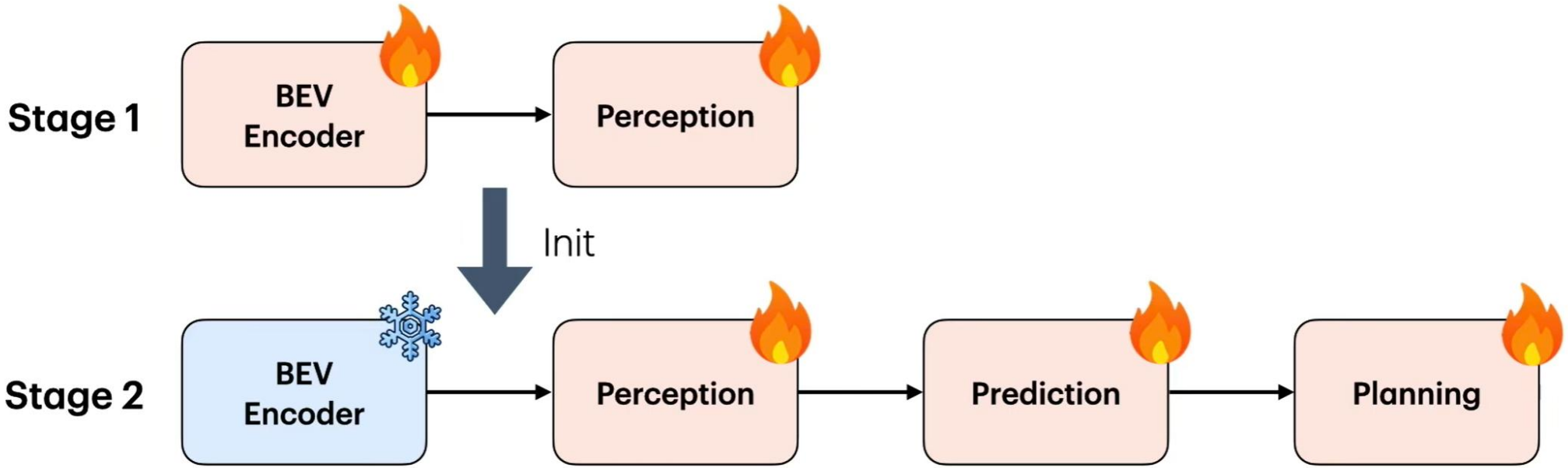
UniAD

- Size of each component

Component	Parameters (Million)	Proportion	Notes
Backbone + BEV Encoder	77–82	~62–65%	Image backbone + BEV feature construction
TrackFormer	8–10	~7%	Agent detection and tracking
MapFormer	8–10	~7%	HD map element segmentation
MotionFormer	10–12	~9%	Multi-agent future trajectory prediction
OccFormer	6–8	~5%	Dense BEV occupancy prediction
Planner	2–3	~2%	Ego vehicle trajectory planning
Total	≈125	100%	Official reported total parameters

UniAD

- Training strategy

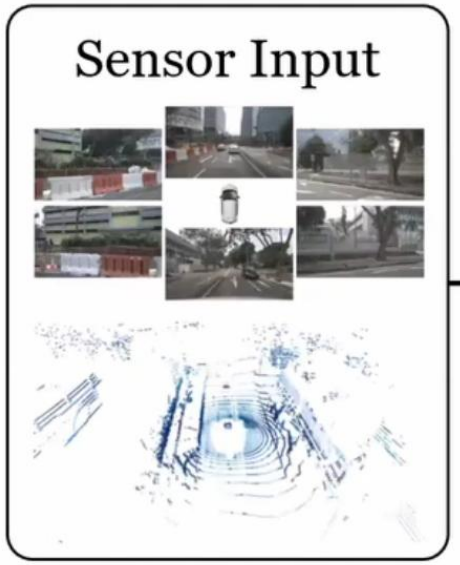


Towards Interpretability

VLM-powered E2E

Towards Interpretability

VLM-powered E2E



You are a driver driving a passenger car on the road. The image shows what you see from the front-facing camera mounted to the vehicle. Provide concise answers to the following questions:

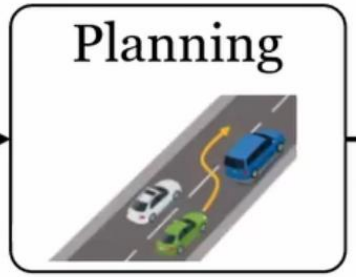
1. Describe what you see
2. Is the road blocked ahead
3. What's the safe future driving action

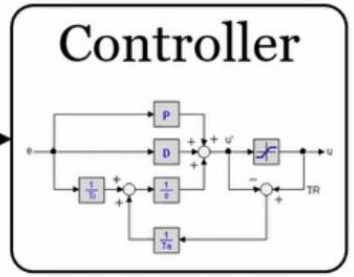
Reasoned about road conditions and actions for 5 seconds

1. There appears to be a large, light-colored bag or sack-like object lying in the middle of the road.

2. The road is partially obstructed, but not fully blocked.

3. Slow down, proceed with caution, and pass around the obstacle safely when clear to do so.



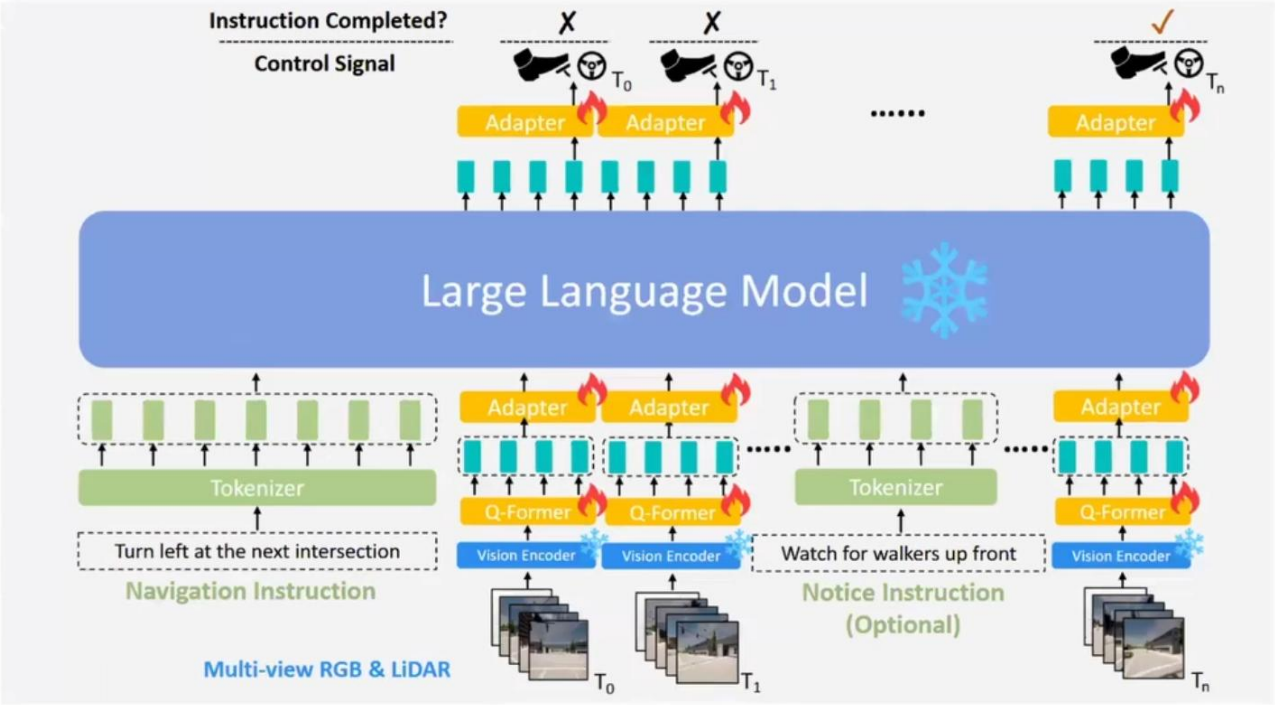


Formulate driving as a VQA task!

Towards Interpretability

VLM-powered E2E

1. Driving with VLM



Architecture of **LMDrive**

✗ Advanced reasoning
✗ Interpretable

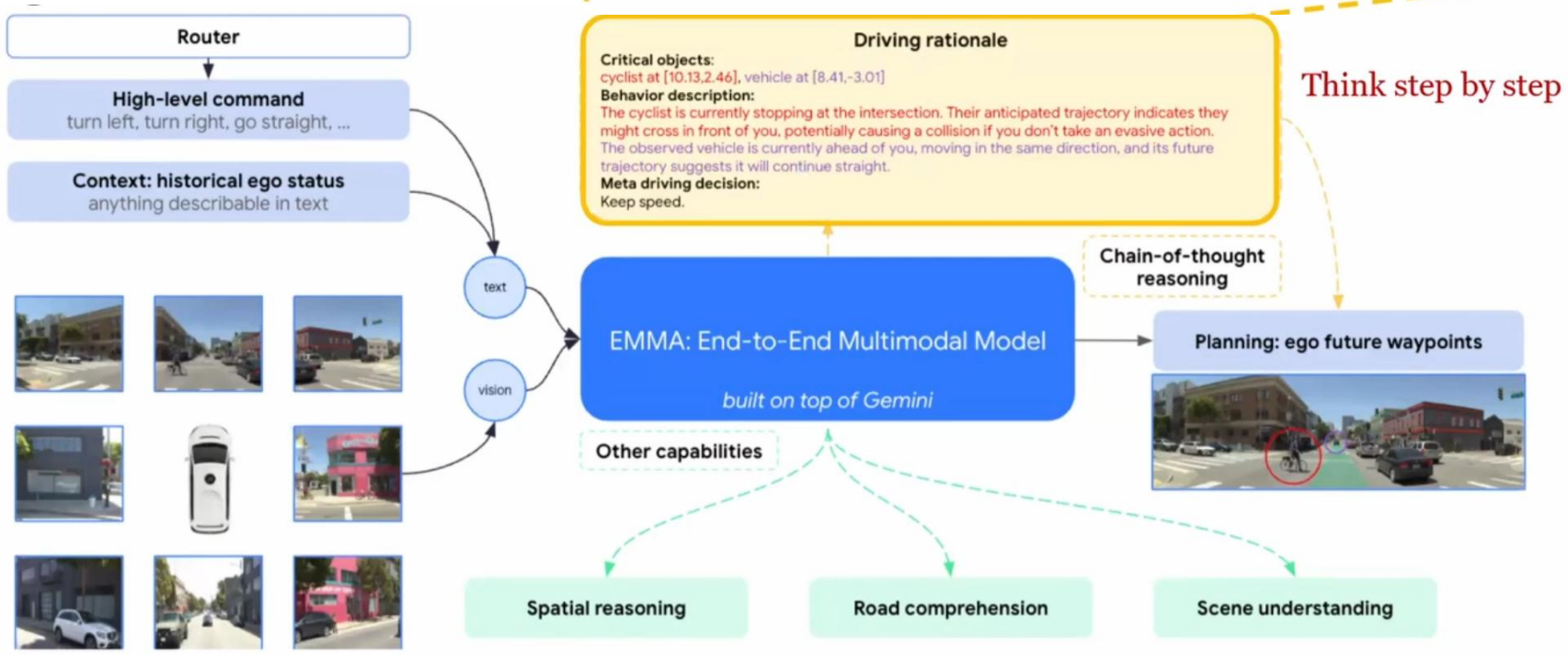
H. Shao et al., "LMDrive: Closed-Loop End-to-End Driving with Large Language Models," CVPR, 2024.

Towards Interpretability

VLM-powered E2E

1. Driving with VLM

Scene description	Critical object	Meta decision	Behavior description	Relative improvements over baseline e2e planning
✓	✗	✗	✗	+ 0.0%
✗	✓	✗	✗	+ 1.5%
✗	✗	✓	✗	+ 3.0%
✗	✓	✓	✗	+ 5.7%
✗	✓	✓	✓	+ 6.7%



Architecture of **EMMA**

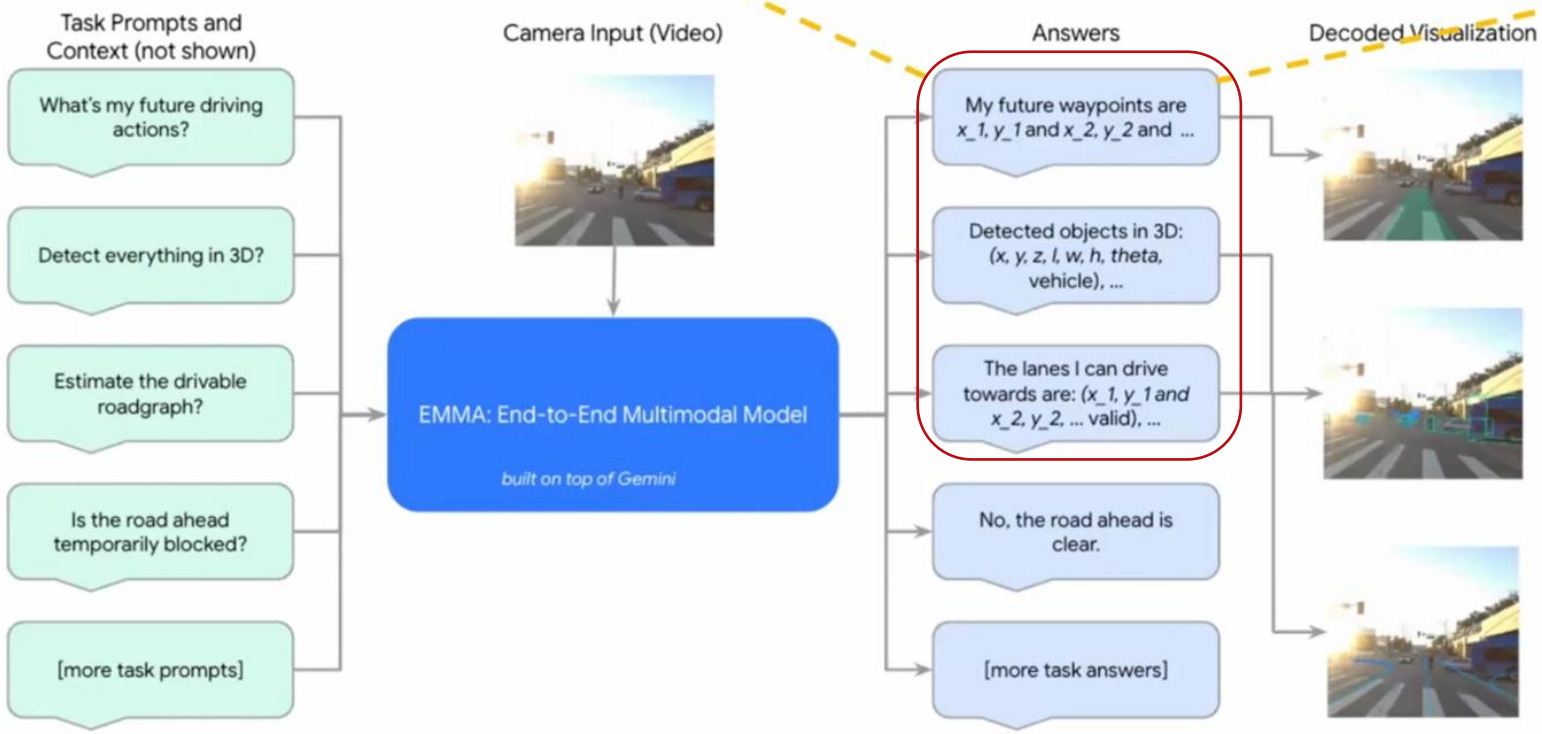
J.-J. Hwang et al., "EMMA: End-to-End Multimodal Model for Autonomous Driving," Waymo, 2024.

Towards Interpretability

VLM-powered E2E

1. Driving with VLM

e2e planning	3D detection	road graph estimation	Relative improvement over single task: e2e planning	detection	road graph
	✓	✓	-	+1.6% (±1.0%)	+2.4% (±0.8%)
✓	✓		+1.4% (±2.8%)	+5.6% (±1.1%)	-
✓		✓	−1.4% (±2.9%)	-	+3.5% (±0.9%)
✓	✓	✓	+1.4% (±2.8%)	+5.5% (±1.1%)	+2.4% (±0.8%)



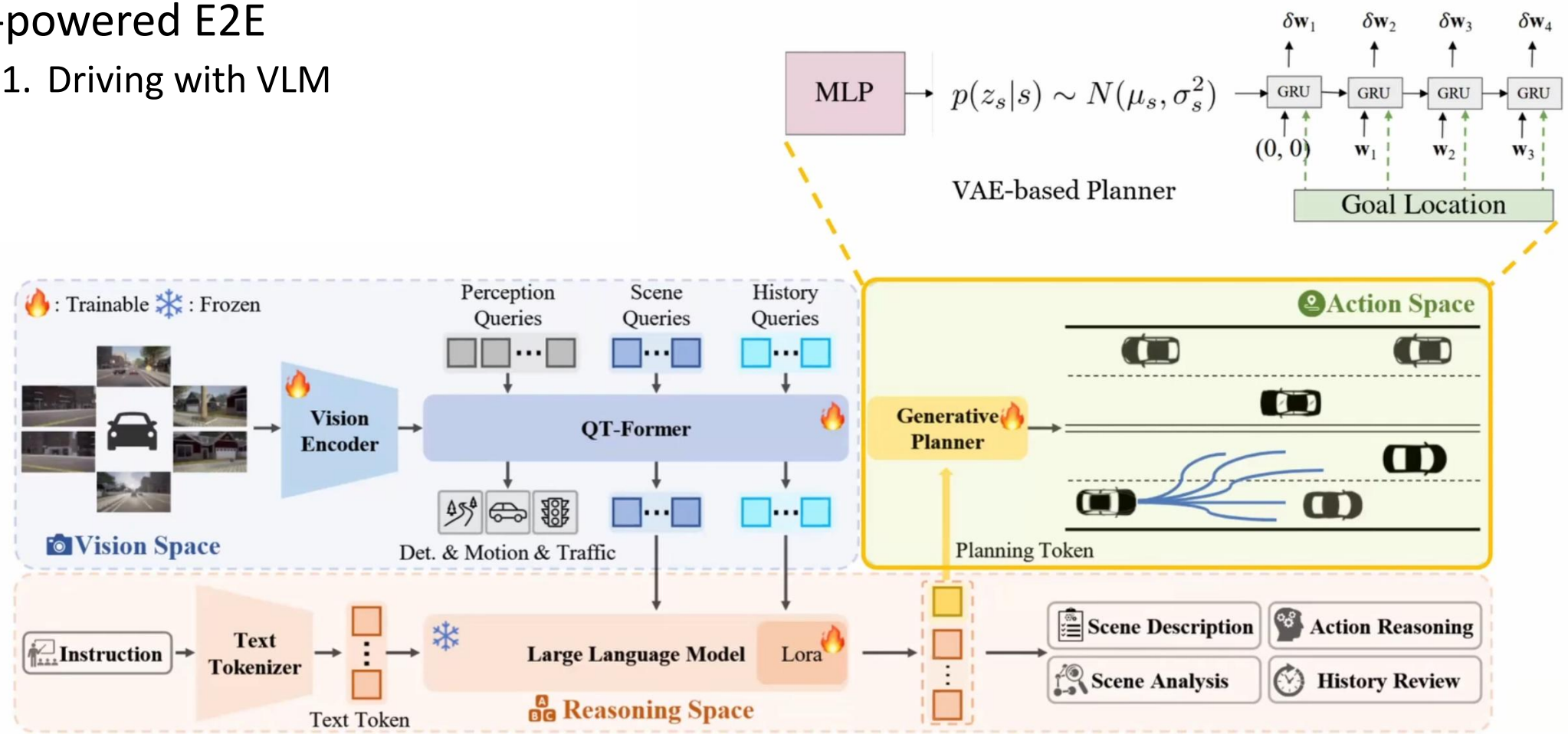
✗ Numeric accuracy
✗ Realistic guarantee

J.-J. Hwang et al., "EMMA: End-to-End Multimodal Model for Autonomous Driving," Waymo, 2024.

Towards Interpretability

VLM-powered E2E

1. Driving with VLM



hOlistic E2E autonomous dRiving framework by vIsion-language instructed actiON generation (ORION)

H. Fu et al., "ORION: A Holistic End-to-End Autonomous Driving Framework by Vision-Language Instructed Action Generation," 2025.

Towards Interpretability

VLM-powered E2E

1. Driving with VLM
   - DriveMLM (Dec 2023)
   - LMDrive (CVPR’24)
   - DriveGPT4 (RA-L’24)
   - DriveVLM (CoRL’24)
   - OmniDrive (NVIDIA, 2024)
   - EMMA (Waymo, 2024)
   - OpenEMMA (WACV’25)
   - ORION (Mar 2025)

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✗ Runtime efficiency
✗ Safety verification

Reasoned about road conditions and actions for 5 seconds

1. There appears to be a large, light-colored bag or sack-like object lying in the middle of the road.

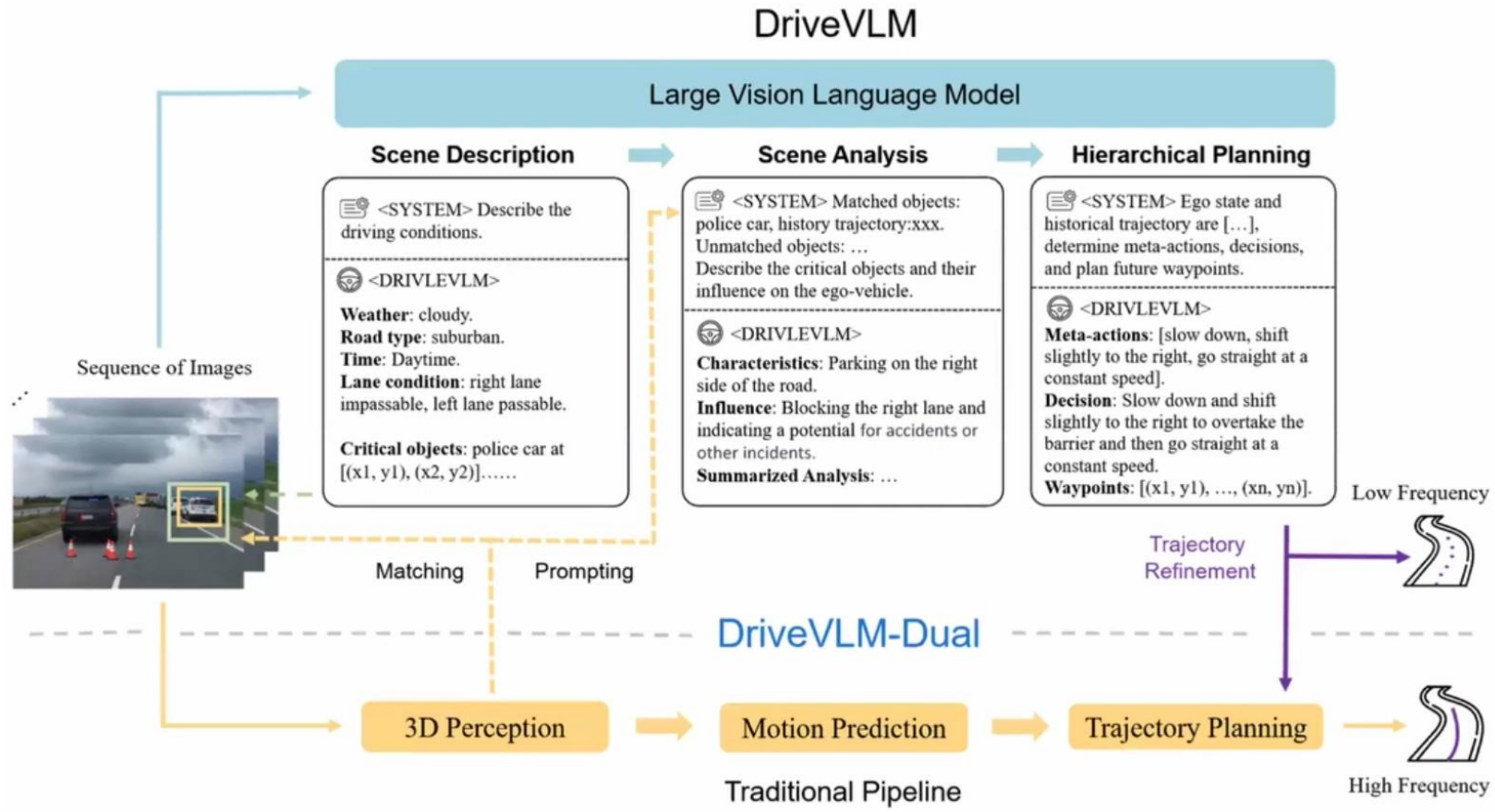
2. The road is partially obstructed, but not fully blocked.

3. Slow down, proceed with caution, and pass around the obstacle safely when clear to do so.

Towards Interpretability

VLM-powered E2E

2. VLM-assisted planning (fast-slow dual system)

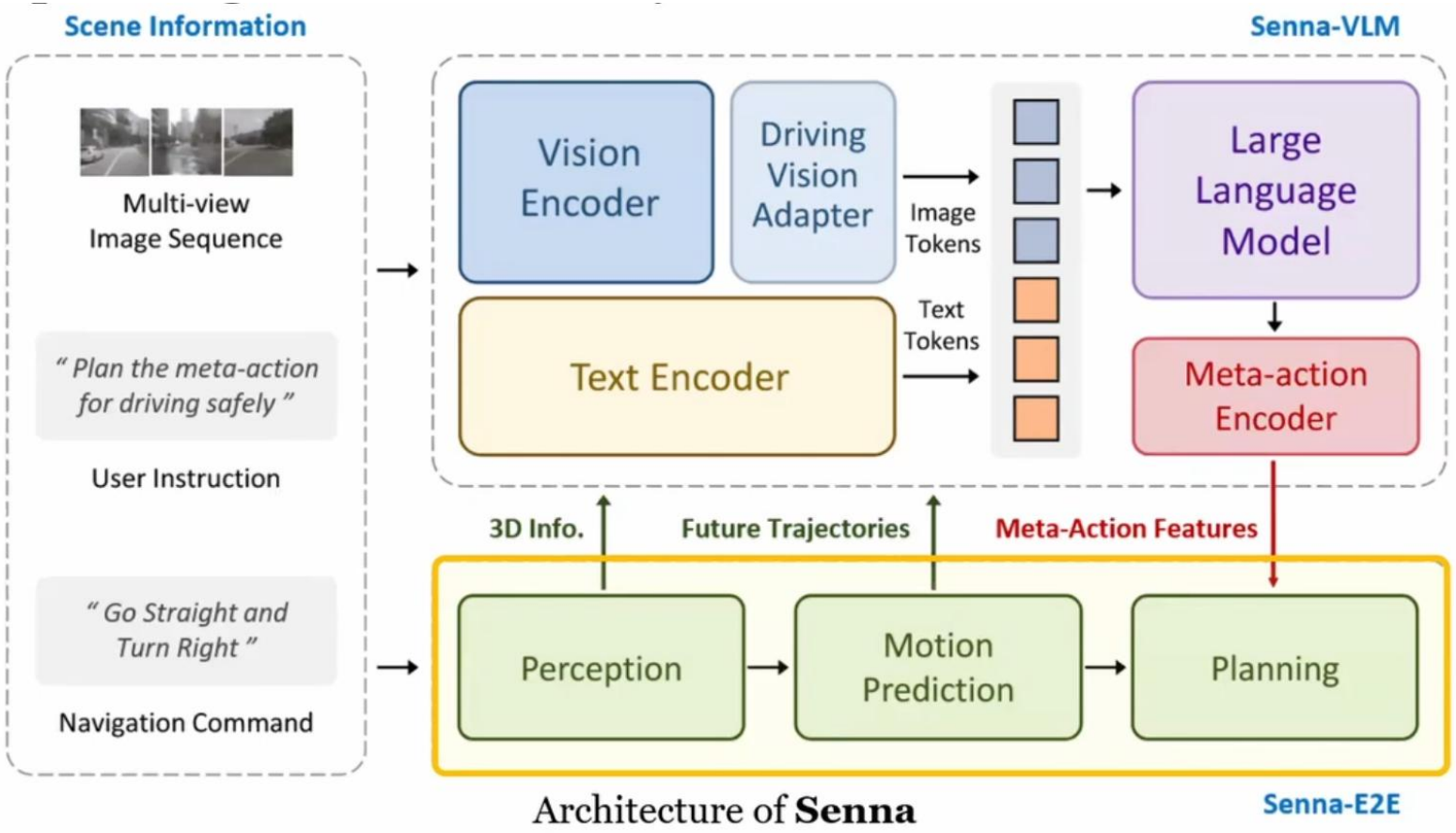


X. Tian et al., "DriveVLM: The Convergence of Autonomous Driving and Large Vision-Language Models," CoRL, 2024.

Towards Interpretability

VLM-powered E2E

2. VLM-assisted planning (fast-slow dual system)



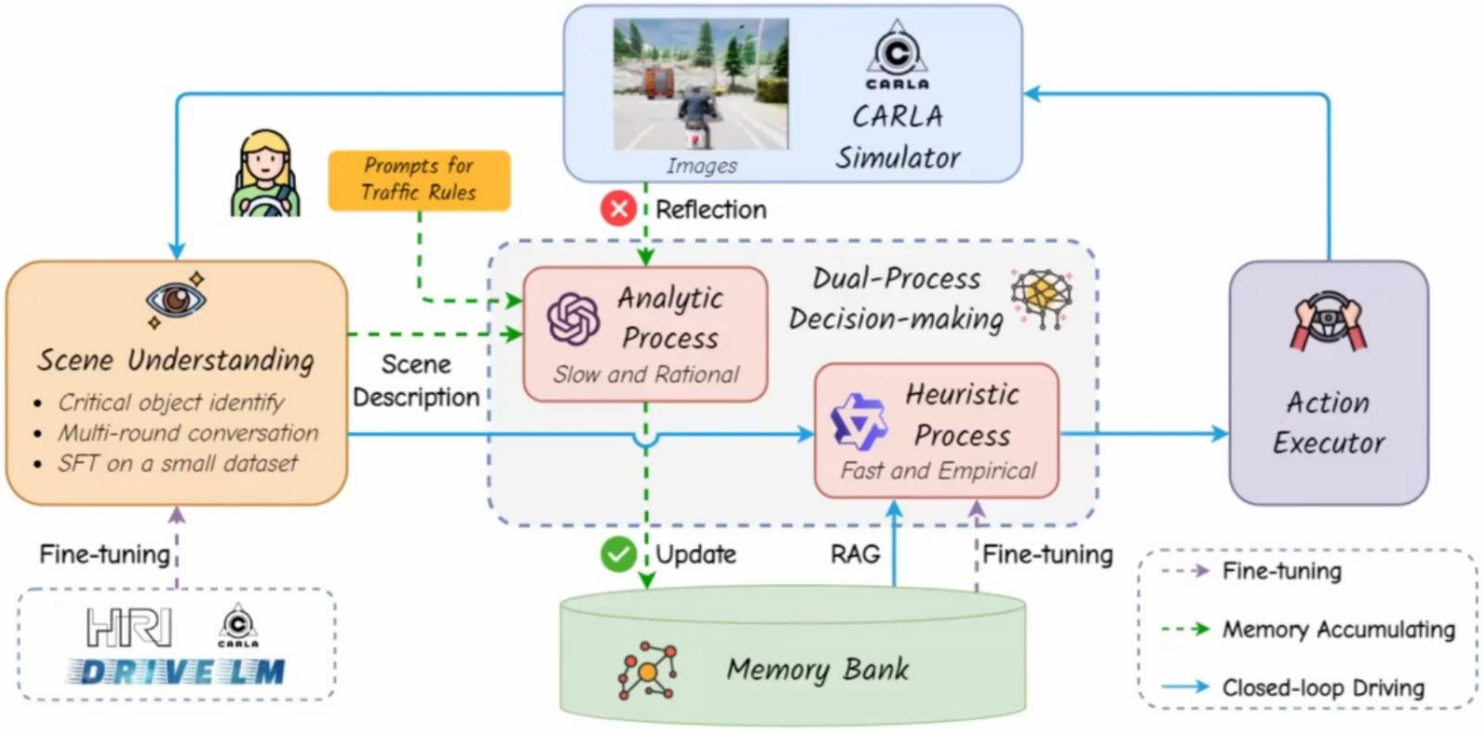
Architecture of **Senna**

B. Jiang et al., "Senna: Bridging Large Vision-Language Models and End-to-End Autonomous Driving," 2024.

Towards Interpretability

VLM-powered E2E

2. VLM-assisted planning (fast-slow dual system)



Architecture of **LeapAD**

J. Mei et al., "Continuously Learning, Adapting, and Improving: A Dual-Process Approach to Autonomous Driving," NeurIPS, 2024.